

PUSHKARAJ BARADKAR

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EDUCATION

University of Southern California, Viterbi School of Engineering GPA: 3.60 / 4.00 Master of Science in Computer Science Machine Learning for Data Science, Analysis of Algorithms, Web Technologies, Database Systems	Jan 2025 - Dec 2026 Los Angeles, CA
University of Mumbai GPA: 9.13 / 10 Bachelor of Engineering in Computer Engineering Data Structures, Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Natural Language Processing	Jul 2020 - May 2024 Mumbai, India

EXPERIENCE

Software Engineer Intern – Vision & Automation USC , Los Angeles, CA • Built a real-time SLAM pipeline combining monocular VO and depth estimation, maintaining 20–25 FPS on indoor sequences. • Implemented keyframe selection and pose optimization, achieving sub-meter drift (<0.8 m ATE over ~100 m) indoor trajectories. • Integrated depth perception for map reconstruction, improving spatial consistency and reducing scale drift across long sequences. • Debugged sensor noise, calibration drift, and VO failure cases, improving trajectory stability and reducing tracking dropouts by ~30%. • Collected and validated vision datasets from real-world camera systems, focusing on consistency, temporal alignment, and robustness.	Nov 2025 – Present
Software Engineer Intern – AI & Computer Vision YMT Medical , India • Engineered computer vision pipelines for image quality enhancement and denoising, improving robustness of AI-driven diagnostics. • Trained and fine-tuned YOLOv8n on 2K dermatology images, supporting model validation and performance testing. • Enhanced GPU inference and integrated clinical feedback to enhance model interpretability and reduce latency by 40%. • Designed evaluation pipelines to measure model performance, interpretability, and latency tradeoffs in production-like settings. • Worked closely with domain experts to refine model outputs and reduce failure cases in real clinical data.	Jan 2024 – Aug 2024
Software Engineer Intern – Backend & APIs Technoriya ERP Solution , India • Developed and optimized backend services and APIs using Python and Node.js, supporting scalable data pipelines, system reliability, cloud-based deployment and processing layers for downstream analytics and ML workflows. • Implemented secure authentication and access protocols via Firebase Auth and Firestore rules to ensure data integrity. • Optimized RESTful APIs and Firestore queries to achieve 450 ms response times under 1,000 concurrent requests for ML integration. • Streamlined deployment workflows using Vercel and GitHub Actions to improve delivery consistency and reliability.	Oct 2023 – Dec 2023

PROJECTS

InfoDistill: LLM-based automation & retrieval system - Link • Built retrieval-based AI systems using embeddings and RAG, designing similarity search and ranking pipelines over unstructured data. • Designed an agentic AI pipeline using FastAPI and Hugging Face models (BART, zero-shot) for relevance scoring and summarization, reducing manual research time by ~70%. • Owned, integrated end-to-end development, and automated Google Doc generation and email dispatch using Arcade SDK.	Jun 2025 – Aug 2025
CampusConnect USC – Full-Stack Web Platform • Developed a web app for verified USC students to post, chat, and review professors using Next.js, Prisma, Supabase, and Google OAuth. • Launched a responsive Google Cloud platform with Supabase backend, enabling anonymous posts, discussions, and professor ratings. • Integrated domain-restricted Google sign-in and GraphQL API supporting student posts and reviews.	Nov 2025 – Present
Artsy App: Full Stack App for Art Discovery and Curation - Link • Engineered a responsive React frontend with optimized rendering, intuitive UI, and integrated rate limiting for stable performance. • Constructed a RESTful API with 5 endpoints using Node.js, Express.js, and MongoDB, managing authentication for 500+ artworks. • Released backend services on Google Cloud App Engine, ensuring scalability and seamless integration with the frontend.	Feb 2025 – March 2025
PayPulse: Full-Stack Peer-to-Peer Payment Platform - Link • Implemented a full-stack peer-to-peer payment platform using React.js, Node.js, and MongoDB, achieving sub-600 ms transaction latency. • Integrated Stripe API with idempotent retries and transactional locks to ensure 99.9% payment reliability. • Implemented JWT-based authentication with AES-encrypted sessions, enhancing security and reducing login errors.	Oct 2024 – Dec 2024

TECHNICAL SKILLS, PUBLICATIONS, CONFERENCES

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript, R, C#, PHP
- **AI / ML & Data Science:** Deep Learning, Generative AI, NLP, LLMs (OpenAI/GPT, Llama), Transformers, RAG, Agentic AI Systems, Autonomous Agents, Prompt Engineering, Computer Vision, Data Mining, Data Science, PyTorch, TensorFlow, LangChain
- **Frameworks & Development:** FastAPI, Flask, Django, Node.js, Express.js, React.js, Next.js, Angular.js, OAuth, OOP, Agile, .NET
- **Databases & Cloud:** MongoDB, SQL, PostgreSQL, Supabase, Redis, AWS, GCP, Azure, Docker, Kubernetes, Git, CI/CD, Gitlab, Linux environments (development, debugging, containerized services)
- **Software Engineering:** Object-Oriented Programming (OOP), design patterns, backend development, REST APIs, debugging, Agile, REST API, AWS, Lambda, EC2, Fargate, CloudWatch, SQS, SNS, SES, SageMaker